

A Level Computer Science (9618)

Topical Past Paper Worksheet

Topic 4: Processor Fundamentals

Subtopic 4.2: Assembly Language

Concept 4.2.2: Two-Pass Assembler

Total Questions: 3

Mark Schemes begin on Page 5

Question 1 | Source: 9618_w23_qp_11 (Q8.a)

8 (a) Identify the purpose of the first pass of a two-pass assembler.

.....

..... [1]

Question 2 | Source: 9618_s23_qp_12 (Q3.a)

3 A program is written in assembly language.

(a) The program is converted into machine code by a two-pass assembler.

Draw **one or more** lines to identify the pass or passes in which each action takes place.

Action	Pass
generates object code	
reads the source code one line at a time	first
removes white space	second
adds labels to the symbol table	

[3]

Question 3 | Source: 9618_w22_qp_11 (Q6.c)

- 6 The following table shows part of the instruction set for a processor. The processor has one general purpose register, the Accumulator (ACC), and an Index Register (IX).

Instruction		Explanation
Opcode	Operand	
LDM	#n	Immediate addressing. Load the number n to ACC
LDD	<address>	Direct addressing. Load the contents of the location at the given address to ACC
LDX	<address>	Indexed addressing. Form the address from <address> + the contents of the index register. Copy the contents of this calculated address to ACC
LDR	#n	Immediate addressing. Load the number n to IX
STO	<address>	Store contents of ACC at the given address
ADD	<address>	Add the contents of the given address to the ACC
ADD	#n	Add the denary number n to the ACC
INC	<register>	Add 1 to the contents of the register (ACC or IX)
CMP	#n	Compare the contents of ACC with number n
JPE	<address>	Following a compare instruction, jump to <address> if the compare was True
JPN	<address>	Following a compare instruction, jump to <address> if the compare was False
OUT		Output to the screen the character whose ASCII value is stored in ACC
END		Return control to the operating system
<address> can be an absolute or a symbolic address # denotes a denary number, e.g. #123 B denotes a binary number, e.g. B01001101		

- (c) Tick (✓) **one or more** boxes in each row to indicate whether the task is performed in the first pass or the second pass of a two-pass assembler.

Task	First pass	Second pass
Remove comments.		
Read the assembly language program one line at a time.		
Generate the object code.		
Check the opcode is in the instruction set.		

[2]

MARK SCHEMES

Mark Scheme for Question 1 | Source: 9618_w23_ms_11 (Q8.a)

8(a)	1 mark for: To create a symbol table	1
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Mark Scheme for Question 2 | Source: 9618_s23_ms_12 (Q3.a)

3(a)	1 mark for generates object code to second pass 1 mark for reads source code one line at a time to both boxes 1 mark for removes white space and adds labels to first pass Action Pass generates object code reads the source code one line at a time removes white space adds labels to the symbol table first second <pre> graph LR A1[generates object code] --> P2[second] A2[reads the source code one line at a time] --> P1[first] A2 --> P2[second] A3[removes white space] --> P1[first] A4[adds labels to the symbol table] --> P1[first] </pre>	3
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Mark Scheme for Question 3 | Source: 9618_w22_ms_11 (Q6.c)

6(c)

1 mark for each pair of highlighted rows

2

Task	First pass	Second pass
Remove comments.	✓	
Read the assembly language program one line at a time.	✓	✓
Generate the object code.		✓
Check the opcode is in the instruction set.	✓	